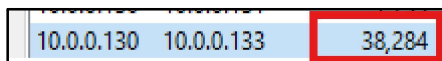


PsExec Lab Walkthrough

An alert from the Intrusion Detection System (IDS) flagged suspicious lateral movement activity involving PsExec. This indicates potential unauthorized access and movement across the network. As a SOC Analyst, your task is to investigate the provided PCAP file to trace the attacker's activities. Identify their entry point, the machines targeted, the extent of the breach, and any critical indicators that reveal their tactics and objectives within the compromised environment.

Q1. To effectively trace the attacker's activities within our network, can you identify the IP address of the machine from which the attacker initially gained access?



10.0.0.130	10.0.0.133	38,284
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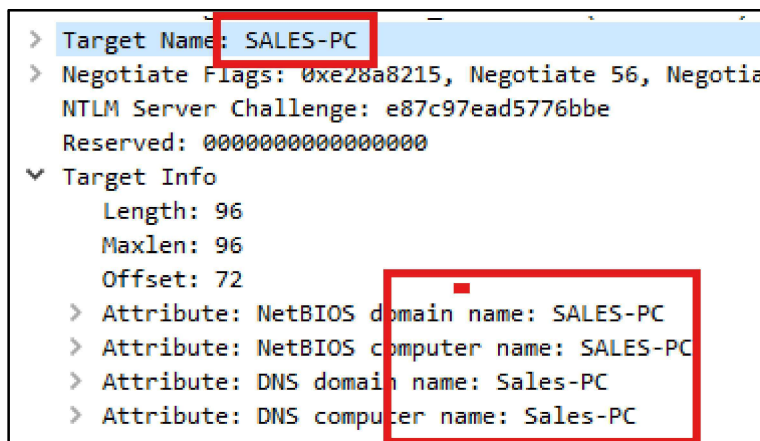
Go check statistic, conversations.

Because it has the most packet meaning there must be a suspicious activity going on between those ip address

10.0.0.130

Q2. To fully understand the extent of the breach, can you determine the machine's hostname to which the attacker first pivoted?

Filter wireshark with attackers ip, then go through the SMB2 packet, in details you will see the hostname the attacker first pivoted.



SALES-PC

Q3. Knowing the username of the account the attacker used for authentication will give us insights into the extent of the breach. What is the username utilized by the attacker for authentication?

```
220 Session Setup Request, NTLMSSP_NEGOTIATE
329 Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP_CHALLENGE
595 Session Setup Request, NTLMSSP_AUTH, User: \ssales
```

Through the packet detail you can see the username utilized by the attacker for auth.

ssales

Q4. After figuring out how the attacker moved within our network, we need to know what they did on the target machine. What's the name of the service executable the attacker set up on the target?

```
10.0.0.133      SMB2      382 Create Request File: PSEXESVC.exe
10.0.0.130      SMB2      410 Create Response File: PSEXESVC.exe
```

Next go deeper you'll see the attacker set up a file into the SMB server. Also the file is .exe which is executable.

PSEXESVC.exe

Q5. We need to know how the attacker installed the service on the compromised machine to understand the attacker's lateral movement tactics. This can help identify other affected systems. Which network share was used by PsExec to install the service on the target machine?

```
▼ SMB2 (Server Message Block Protocol version 2)
  ▼ SMB2 Header
    ProtocolId: 0xfe534d42
    Header Length: 64
    Credit Charge: 1
    Channel Sequence: 0
    Reserved: 0000
    Command: Create (5)
    Credits requested: 1
    > Flags: 0x00000030, Priority
    Chain Offset: 0x00000000
    Message ID: 9
    Reserved: 0x0000feff
    > Tree Id: 0x00000005 \\10.0.0.13 \ADMIN$
    > Session Id: 0x000030000000000039 Acct:ssales Domain: Host:HR-PC
    Signature: 00000000000000000000000000000000
    [Response in: 145]
  ▼ Create Request (0x05)
    > StructureSize: 0x0039
```

Check packet detail SMB2, show that network share used was ADMIN

ADMIN\$

Q6. We must identify the network share used to communicate between the two machines. Which network share did PsExec use for communication?

```
159 Session Setup Response
164 Tree Connect Request Tree: \\10.0.0.133\IPC$
138 Tree Connect Response
178 Ioctl Request FSCTL_QUERY_NETWORK_INTERFACE_INFO
474 Ioctl Response FSCTL_QUERY_NETWORK_INTERFACE_INFO
168 Tree Connect Request Tree: \\10.0.0.133\ADMIN$
```

The attacker uses machine IPC\$ to communicate with ADMIN\$.

IPC\$

Q7. Now that we have a clearer picture of the attacker's activities on the compromised machine, it's important to identify any further lateral movement. What is the hostname of the second machine the attacker targeted to pivot within our network?

Use this filter = ntlmssp.messageType == 2

```
Maxlen: 128
Offset: 80
> Attribute: NetBIOS domain name: MARKETING-PC
> Attribute: NetBIOS computer name: MARKETING-PC
> Attribute: DNS domain name: Marketing-PC
> Attribute: DNS computer name: Marketing-PC
> Attribute: Timestamp
> Attribute: End of list
> Version 10.0 (Build 19041); NTLM Current Revision 15
```

This field can help identify the hostname or domain presented by the SMB server during NTLM authentication.

messageType == 2 , Show the packet where the server sends an NTLM challenge to the client.